## Machine Learning Driven Assessment of Global Life Expectancy Trends

## Motivation

**Why are we doing your project?**
The global landscape of human health is shaped by a complex interplay of economic, social, and clinical factors. Our project seeks to analyze a comprehensive dataset spanning 193 countries from 2000 to 2015 to uncover the primary drivers of human longevity. By examining diverse variables—ranging from economic indicators like **GDP** and **percentage expenditure** on health to clinical factors such as **immunization coverage (Hepatitis B, Polio, Diphtheria)** and disease prevalence (**HIV/AIDS**) —we aim to provide a data-driven perspective on global health trends. Personally, we are motivated by the potential to use machine learning to identify which specific interventions, such as increasing **schooling** or reducing **adult mortality**, have the most significant statistical impact on life expectancy.

**What is the goal of our application?**
The goal is to develop a robust predictive model that estimates **Life Expectancy** based on a country's socioeconomic and health profile. Beyond simple prediction, the application aims to rank the importance of various features—such as **Income composition of resources**, **BMI**, and **Alcohol consumption**—to determine which factors most reliably predict a longer life.

**Why is it important?**
This research is critical for global policy-making and resource allocation. Understanding the relationship between **total health expenditure** and actual health outcomes (like reduced **infant and under-five mortality**) allows governments and international organizations to prioritize investments. For instance, determining whether **schooling** has a stronger correlation with life expectancy than direct health spending can shift the focus of long-term developmental strategies.

**What is our hypothesis?**
Our hypothesis is that life expectancy is a highly predictable outcome when modeled against a combination of economic stability and public health infrastructure. Specifically, we expect that **Income composition of resources** and **Schooling** will be the strongest positive predictors, while **HIV/AIDS** prevalence and **Adult Mortality** will act as the primary negative constraints. We believe the problem is solvable because the dataset provides a consistent 15-year timeline that captures the measurable impact of global health initiatives and economic shifts across diverse national "Statuses" (Developing vs. Developed).

## Learning task

**Where is the data from?**
Our data comes from the World Health Organization (WHO). It tracks health and economic factors for 193 countries between the years 2000 and 2015.

**Why can this data solve the problem?**
This data is useful because it contains the "ingredients" that usually determine how long people live. It includes 20 different factors like wealth (GDP), education (Schooling), and

health safety (Immunizations). By looking at 15 years of history, the computer can learn the relationship between these factors and the final age people reached.

**Data Examples:**

- **Example 1:** A country with high **Schooling** (e.g., 15 years) and high **GDP** usually shows a **Life Expectancy** of 75+ years.
- **Example 2:** A country with high **Adult Mortality** and high **HIV/AIDS** rates usually shows a lower **Life Expectancy** of around 50–60 years. The computer uses these patterns to understand that when "Factor A" goes up, "Life Expectancy" usually goes up or down.

## Learning definition

**What are we going to do?**

We are going to build a system that looks at a country's stats (like its BMI rates, alcohol consumption, and polio vaccine coverage, GDP) and predicts the average age a person there will live to.

**Which learning type is suitable?**

This is a **Supervised Learning** task, specifically **Regression**.

- It is "Supervised" because we already have the correct answers (the actual life expectancy) to teach the computer.
- It is "Regression" because we are predicting a specific number (years of age), not just putting things into categories.

## Performance Measure

**How will we measure success?**

We will use **Mean Absolute Error (MAE)**. This measure tells us, on average, how many "years" our prediction was away from the real answer.

**Why this measure?**

MAE is very easy to understand in the context of human life. If our MAE is 2.5, it means our typical prediction is off by two and a half years.

**How to interpret the results?**

- **MAE = 1.0 to 2.0 (Excellent):** If our model is only off by 1 or 2 years, it is very accurate and can be used to help governments make real decisions.
- **MAE = 5.0 or higher (Poor):** If the model is off by 5 or 10 years, it is not very useful because that is a huge difference in terms of a human's lifespan.

## Analysis of Previous Work

**Problem:** The study addresses the difficulty physicians face in accurately forecasting life expectancy, which often leads to missed opportunities for timely medical care or therapy planning. While prior research has looked at demographics and death rates, this paper specifically highlights that the impact of **immunizations** (Polio, Hepatitis B, Diphtheria) on standards of living was previously under-considered.

**Suggested Approach:** The researchers utilized the same **WHO dataset from Kaggle** that you are using, covering a 15-year period. They tested multiple machine learning regression algorithms, including **Logistic Regression, SVM, Decision Tree, and Random Forest Regression**.

**Obtained Results:** The study achieved a strong R-squared value, specifically noting that the **Random Forest algorithm** provided the most accurate results for life expectancy prediction. **How to apply this to your problem:** You can adopt their focus on "Immunization features" as key predictors in your own models to see if they hold the same weight in your specific analysis.

**Required Modifications:** While the paper mentions Logistic Regression, since your task is a regression problem (predicting a continuous age), you should ensure you are using the regression version of these algorithms rather than the classification versions often associated with those names.

**Use as a Baseline:** This paper serves as an excellent baseline for your group's **Random Forest** model (Student B). You can compare your R-squared and error values directly against their findings to evaluate your model's performance.

## Plan

To solve this task, our group will implement three distinct machine learning pipelines to compare performance and ensure "significantly different" methods:

- **Gulkhayo Urinova (Linear Approach):** Will focus on **Multiple Linear Regression** with a focus on feature selection (e.g., using LASSO to handle multicollinearity between variables like GDP and Schooling).
- **Mert Baran Demirtas (Ensemble Approach):** Will implement a **Random Forest Regressor**, focusing on non-linear relationships and feature importance rankings to see which health factors are "deal-breakers" for longevity.
- **Yalin Ege Dalcan (Boosting/Non-Linear Approach):** Will utilize **XGBoost or Support Vector Regression (SVR)**, focusing on hyperparameter optimization (loss functions and learning rates) to minimize the impact of outliers in developing nations.

**Implementation Details:** We will use Python's `scikit-learn` for the core models. Preprocessing will include handling missing values (imputation) and normalizing features like Population and GDP which exist on vastly different scales.

## Risk Management

If our complex models (like XGBoost) don't "converge" (fail to find a solution) or if they "overfit" (become too specific to the training data), we will:

- **Simplify:** Switch to a simpler Linear Regression model.
- **Regularization:** Use "Penalty" techniques (Lasso) to force the model to be less complex.
- **Cross-Validation:** Use K-Fold testing to ensure the model works on different slices of the data.

## Links

- https://www.researchgate.net/publication/364430951_Life_Expectancy_Prediction_through_Analysis_of_Immunization_and_HDI_Factors_using_Machine_Learning_Regression_Algorithms
- https://ir.uitm.edu.my/id/eprint/113319/1/113319.pdf
- https://arxiv.org/pdf/2510.00542
- https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who